

PCM Channels

1. Mono PCM (1 Channel)

- A single audio channel. All sound elements are combined into one signal.
- Sound comes from a single point or sounds identical out of both the left and right speakers/earbuds. There is no spatial depth.
- Samples are stored sequentially for that single channel: [sample1 sample2 sample3 ...]
- A telephone call, an old AM radio broadcast, or a single microphone recording a voice memo.

2. Stereo PCM (2 Channels)

- Stereo PCM uses **two independent audio channels**: Left (L) and Right (R).
- It creates a **2D spatial soundstage**, allowing sounds to be positioned between the left and right speakers. For example, a guitar can be heard more strongly on the left while a drum set is heard more strongly on the right.
- **Interleaved Stereo**

In interleaved stereo, samples from the Left and Right channels are stored alternately in memory: [left1 right1 left2 right2 left3 right3 ...]

This means that each Left sample is immediately followed by the corresponding Right sample.

- **Non-Interleaved Stereo (Planar Stereo)**

In non-interleaved stereo, the samples of each channel are stored in separate, continuous blocks: [left1 left2 left3 ... leftn] [right1 right2 right3 ... rightn]

Both formats contain the same audio information; they only differ in **how the samples are arranged in memory**.

- For example, Spotify music, modern film audio, or music listened to through headphones can use stereo audio, where instruments are positioned across the left and right channels

More channels = more spatial information and immersion, but also more data and storage.

Fewer channels = less storage and lower data rate, but less spatial detail and immersion.